

Timothy Singh

Data Scientist | Data Analyst

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Skills

- **Programming:** Python, Java, C++, JavaScript, SQL, C#, Bash
- **Web & Backend:** Spring Boot, Flask, Node.js, Express.js, REST APIs, HTML5, CSS3
- **Machine Learning & AI:** scikit-learn, PyTorch, Ensemble Models, NLP (spaCy, NLTK, BERT), CNNs, RNNs, Transformers, SHAP
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, seaborn, ggplot2, Altair, Dash, Shiny, Power BI
- **Statistics:** Regression, Hypothesis Testing, A/B Testing, Time Series (ARIMA, LSTM)
- **Cloud & DevOps:** AWS (S3, Lambda, EC2), Azure, Docker, Git, CI/CD (GitHub Actions), Linux
- **Big Data & Databases:** Hadoop, Spark, MapReduce, SQL/NoSQL (Oracle, MySQL, PostgreSQL, MongoDB)
- **Tools:** Jupyter, NetBeans, VS Code, IntelliJ, Eclipse, Snowflake, Microsoft Azure

Education

M.Sc. Data Science (Average: 95.0)

University of British Columbia, Vancouver, 2025

Completed courses in inferential statistical analysis, supervised and unsupervised learning, data structures, data visualization, databases, spatial and temporal models, A/B testing, Bayesian/Frequentist regression, cloud computing (AWS).

B.Sc. Computer Science (Special) – First Class Honours (GPA: 4.23)

University of the West Indies, St. Augustine Campus, 2021

Completed courses in procedural and object-oriented programming, networking, cyber-security, algorithms, data structures, machine learning/intelligent systems and data analytics.

Career Experience

DataAnnotation, Remote

July 2025 – present

Data Annotator

- **Evaluated and ranked** ML model outputs across text, reasoning, and code tasks by **designing** standardized scoring rubrics, **improving consistency and reliability** of performance assessments.
- **Delivered structured qualitative feedback** including code correctness, efficiency, edge-case handling, and instruction adherence, **directly supporting model training and refinement**.

ALS Geoanalytics, North Vancouver, British Columbia, Canada

April 2025 – July 2025

Data Science Intern (Capstone Project)

- **Designed, developed and benchmarked a Python package** to measure and track carbon emissions of large-scale deep learning models, **achieving 85% measurement accuracy** compared to existing packages.
- **Integrated the framework with any Python workflow**, enabling seamless adoption of this package to **any PyTorch workflow**.
- **Leveraged real-time carbon intensity APIs** to provide **hourly** regional emissions updates, **ensuring 100% up-to-date tracking**.
- **Enabled tracking of carbon emission reduction techniques** that reduced the carbon footprint of model training by **up to 20%**, **using pruning and quantization**, supporting company sustainability initiatives.

Ministry of Digital Transformation (MDT) of Trinidad and Tobago, Port of Spain, Trinidad and Tobago

September 2021 – August 2024

Digital Transformation Analyst – ICT Services Division

- **Optimized inter-Ministry request and approval processes** by developing a centralized online platform, reducing request processing time by **30%** and increasing request visibility across **8 ministries**.
- **Engineered and deployed ICT Access Centres** in **8 communities**, providing underserved areas with access to cyber cafes and training rooms, impacting over **1,500 residents**.
- **Configured, deployed, and maintained** Ubuntu 20.04 environments for 200+ staff using **Bash scripting** and **network imaging tools**, while executing **bug tracking** and **unit-level testing** to ensure reliability, resulting in a 40% decrease in on-boarding times.
- **Implemented an open-source Asset Management system with LDAP-AD integration**, automating asset tracking and improving deployment efficiency by **35%**.
- **Supported data cleaning and wrangling** of budget spreadsheets to assist in fund allocation for fiscal year **2022–2023**.
- **Managed and resolved** Tier 0–3 technical support requests using **ticketing software** and **remote troubleshooting tools** (AnyDesk, Microsoft Teams), handling 100+ issues monthly and independently resolving **85%**.
- **Developed the Ministry's first global website**, increasing domestic and international exposure by **90%**.

- **Engaged in procurement activities** for ICT and office equipment (servers/client machines), improving the Ministry's technical working environment.

Republic Bank Limited, Information Technology Management Division, Trincity, Trinidad and Tobago

June 2020 – August 2020

Intern – Credit Card Support Division

- **Cleaned and transformed** large, complex **Excel datasets** using **pandas (Python)** to prepare structured inputs for ingestion into an internal **SQL database**, **improving data accessibility** for the analytics team.
- **Designed and built** a secure, front-end **user interface** using **Apache Netbeans** that enabled **credit card administrators** to manage database entries (**create, edit, delete**) without backend access, **reducing administrative overhead by 70%**.

Key Projects

- **Database GUI** - This system was developed during internship at Republic Bank Limited. It was created with **Apache Netbeans using Java**. This system allowed for simplified access to an AS400 database system containing data for **1500+ employees**, allowing for CRUD operation using **Java Database Connectivity (JDBC)**.
- [**dsresumatch**](#) - DSResuMatch is a package designed to analyze resumes and job descriptions by extracting key sections and scoring them based on keyword relevance.
- [**NIH Chest X-Ray Classifier**](#) - This was part of a Hack-a-thon in which a **classifier model** was trained using thousands of images of lung x-rays, to identify diseases and disorders. Transfer learning was used on an existing DenseNet model and fine-tuned to achieve a decently performing model. Future work would include the inclusion of image transformation and feature engineering.
- [**Credit Card Default Predictor**](#) - This project aims to predict the likelihood of a customer defaulting on their credit card payments using historical payment data. The model incorporates various feature engineering techniques to improve predictive accuracy and reduce overfitting. The dataset was obtained from [**Kaggle**](#) containing around **30000 rows with 25 features**.
- [**Robotic Arm Project**](#) - A project that focused on the programming of a Mecademic500 robotic arm, to pick up blocks and place them at a precise location. This was done in collaboration with [**Virtana**](#), using **Robotic Operating System (ROS)** and **Gazebo** within a **Linux** environment. Presentation: <https://youtu.be/JDTCNuB09mc>
- [**Project Bloodline**](#) - A Progressive Web Application which serves to essentially sensitize and encourage users to donate blood through user-friendly interface. The project was created using the **JAM (JavaScript API Markup)** stack with the front-end application being built with **Angular 9.1.1** and the back-end application server with **Flask 1.0.2**, deployed on Heroku, ensuring **100% uptime**.
- [**Pharmacy POS System**](#) - This was a project carried out as part of the course: Software Engineering II at the University of the West Indies. Utilizing an agile methodology for development, this web application consisted of a remote database of **100+ pharmacy products**, as well as **point of sale capabilities** using **Flask, SQL** and **Angular**.

Certifications

- Microsoft Office Specialist – Microsoft Excel Certification (2021)
- Project Management Fundamentals (2023)
- Project Management Applications (2023)
- IBM - Introduction to Artificial Intelligence (2024)
- IBM – Generative AI: Introduction and Applications (2024)
- IBM – Generative AI: Prompt Engineering Basics (2024)
- Create Machine Learning Models in Microsoft Azure (2024)